

Attention Mechanism in AI

Introduction:

Attention is one of the most powerful concepts in modern Artificial Intelligence, especially in NLP and Deep Learning.

Why Attention is Needed:

Traditional models like RNNs struggle with long sequences. Attention helps focus on important parts.

What is Attention:

It allows models to focus on relevant data instead of all input equally.

Real-Life Example:

Like humans focusing on keywords while reading.

How Attention Works:

Uses Query, Key, and Value to compute importance scores.

Types of Attention:

Self-Attention, Global Attention, Local Attention.

Applications:

Machine Translation, Chatbots, Text Summarization.

Conclusion:

Attention helps machines understand context better and is core to Transformers.